GROUP ACTION

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1. Action

Action of an algebraic object on another is a general construct of which group action is one. The simplest form of an action involves a set X acting on another set Y, since a set is an object with no algebraic structure whatsoever. Such an action is given by a function $A : X \times Y \to Y$. For each $x \in X$, this produces an endofunction $A_x : Y \to Y$. Therefore the image $A(X \times Y)$ of the action is $\bigcup_{x \in X} A_x(Y) \subset Y$. This is exactly why it is called an action *by* X (or X acts *on* Y). Since every $x \in X$ morphs Y into a subset $A_x(Y)$ of itself.

Now one can talk about the map $x \mapsto A_x$ given by the *induced function* $\hat{A} : X \to \text{Fun}(Y)^*$ of the action, using which we specialize actions of other algebraic objects. The specialization criteria 'often' requires the induced function $\hat{A}$ of the action A to be a morphism of the category of the actor; 'often' because of the following things: It's not always possible to give $\text{Fun}(Y)$ an algebraic structure that makes it an object of the category of the actor, which is necessary to make $\hat{A}$ a morphism of the same category. And not all categories are algebraic; **Top** is a category, but a topological space is not an algebraic object. And these are not *all*. So using categorical language may sound like a bad idea, as it takes away the accuracy of the argument, which it clearly does. But here is why it is useful! It gives a unified picture of things that are classically treated separately. Things like *group action* (classically limited to action of groups on sets and groups mainly), *ring representation* (through action of a ring on an ablian group), and *linear representation* (through action of a group on a vector space) can be seen as varied manifestations of the same underlying construct in action, which is *action*.

Definition 1.0.1 (Faithful action)**.** An action $A : X \times Y \to Y$ is faithful if its induced function $\hat{A}$ is injective, and unfaithful otherwise.

Remark 1.0.2. I'm purposefully ignoring the codomain description of $\hat{A}$, as it differs for different actor-actee pairs; specifying it, however, is unnecessary, as the injectivity of $\hat{A}$ is purely a set-theoretic property that depends solely on its domain X.

Algebraic structure of the codomain of $\hat{A}$ (say Z) for some actor-actee pairs:

Category of X	Category of Y	Z	Category of Z	Formal Name
Set	Set	$\text{End}(Y)$	Set	Action
Grp	Set	$\text{Aut}(Y)$	Grp	Group Action
Grp	Vect	$\text{Aut}(Y)$	Grp	Group Representation
Ring	Ab	$\text{End}(Y)$	Ring	Ring Representation
Alg	Vect	$\text{End}(Y)$	Alg	Algebra Representation

The endomorphisms and automorphisms in the third column are of the category of Y. When Y is a set, $\text{End}(Y) = \text{Fun}(Y)$; when Y is a vector space, $\text{Aut}(Y) = \text{GL}(Y)$, the general linear group of

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*$\text{Fun}(Y)$ is the set of all functions from Y to Y.

Y etc. But these are just sets of morphisms, what gives them the algebraic structure as given in the fourth column? There are canonical/natural ways of giving them the above-mentioned algebraic structures, which makes them objects of the same category as X, which is necessary to make $\hat{A}$ a morphism of the same category, which is what we needed to specialize the action A to an action of the algebraic object X. For example, function composition '$\circ$' makes $\mathrm{Aut}(Y)$ a group when a group acts on a set or a vector space. In the case of a ring acting on an abelian group, the same function composition makes $\mathrm{End}(Y)$ a monoid, to which we additionally give an abelian group structure using another operation '$+$' (to make '$(\mathrm{End}(Y), +, \circ)$' a ring), whose pointwise definition exploits the operation of the actee abelian group as follows: For all $f, g \in \mathrm{End}(Y)$, $(f+g)(y) := f(y)+g(y)$ for all $y \in Y$. The operation (addition) on the right-hand side is the operation of the abelian group Y; the operation on the left-hand side is the operation (function addition) on $\mathrm{End}(Y)$ that we defined using the abelian group's operation.

References

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